Unit IV:
Human Resource Management and Control

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HRM is managing people so that work gets done and people still smile at the end of the day.

Human Resource Management

- *Human resources is used to describe both the people who work for a* ***company*** *or* ***organization*** *and the* ***department*** *responsible for managing resources related to employees.*
- *process* ***engages*** *in* ***managing human energies, competencies*** *and* ***potentials to attain the organizational goal****.*
- *HRM* ***balance*** *the growth of* ***both employees as well as organization****.*

"HRM is a **strategic approach** to managing employment relations that emphasizes **leveraging human capabilities for competitive advantage**.

-*Michael Armstrong*

- HRM is the strategic and coherent approach to managing an organization’s most valuable asset
- its people to achieve competitive advantage and organizational goals.

Main Evolution of HRM

- ***Pre-1900s*** → *Labour as commodity*
- ***1900s–1930s*** → *Welfare & Scientific Management*
- ***1930s–1950s*** → *Human Relations Movement*
- ***1950s–1970s*** → *Personnel Management*
- ***1980s*** → *Birth of HRM*
- ***1990s–2000s*** → *Strategic HRM (SHRM)*
- ***2010s–Now*** → *Digital, Agile & People-Centric HRM*

Functions of HRM

- *Planning*
- *Recruiting*
- *Training*
- *Compensating*
- *Appraising*
- *Ensuring employee welfare*

- **Planning**: Developing strategies to meet organizational goals and aligning human resources accordingly.
  – ***Example***: A company forecasts a need for 20 new software developers in the next year due to a new project and creates a hiring plan to recruit them.
- **Recruiting**: Attracting and selecting qualified candidates to fill job vacancies.
  – ***Example***: Using online job boards and social media campaigns to attract applicants for a marketing manager position.
- **Training**: Providing employees with the skills and knowledge necessary to perform their roles effectively.
  – ***Example***: Conducting a workshop on leadership skills for potential team leaders within the organization.

- **Compensating**: Establishing and managing salary structures and benefits to reward employees for their work.
  - ***Example***: Implementing a performance-based bonus system that rewards employees based on their annual performance reviews.
- **Appraising**: Evaluating employee performance to provide feedback and guide career development.
  - ***Example***: Conducting annual performance reviews where managers assess employee achievements and set goals for the next year.
- **Ensuring Employee Welfare**: Promoting a healthy work environment and supporting employee well-being and satisfaction.
  - ***Example***: Offering wellness programs, such as gym memberships or mental health days, to improve employee health and morale.

1. Managerial Functions of HRM

- **Planning**:
  - *Forecast workforce needs, develop HR strategies, analyze labor market trends.*
- **Organizing**:
  - *Structure HR activities, define roles, coordinate with departments.*
- **Directing**:
  - *Motivate employees, implement policies, communicate expectations.*
- **Controlling**:
  - *Monitor HR activities, evaluate outcomes, correct deviations.*

2. Operational Functions of HRM

- **Recruitment and Selection**:
  - *Attract and hire qualified candidates.*
- **Training and Development**:
  - *Provide orientation, training, and career development.*
- **Compensation and Benefits**:
  - *Manage pay structures and benefits.*
- **Performance Management**:
  - *Set standards, conduct appraisals, provide feedback.*
- **Employee Relations**:
  - *Handle grievances, ensure compliance, foster positive environment.*
- **Health and Safety**:
  - *Implement safety programs, promote wellness.*
- **HR Administration**:
  - *Maintain records, manage payroll, ensure legal compliance.*

Job Analysis, Job Specification, Job Description

- **Job Analysis**
  - *Systematic process of collecting, analyzing and documenting all information about a job to determine its duties, responsibilities and required skills.*
- **Job Description (JD)**
  - *A written statement that describes what the job holder does, how it is done, under what conditions, and why the job exists.*
- **Job Specification (JS)**
  - *A written statement that specifies the minimum qualifications, skills, education, experience, physical and mental abilities required by a person to perform the job successfully.*

Job analysis

- Job analysis is a systematic process of gathering, documenting and analyzing information about the responsibilities, tasks, skills, abilities, knowledge area, and work context associated with a particular job.
- It forms the basis for defining the right requirements for successfully performing that job.
- A personnel manager has to undertake job analysis so as to put right man on right job.

- There are two outcomes of job analysis:
  - ***Job description***
  - ***Job specification***

key objectives of job analysis

- identify the core duties and responsibilities that a job entails
- Determine the specialized skills, credentials or competencies needed for the job
- Recognize the key performance indicators to measure outcomes for the job
- Understand the environmental/cultural context and physical demands of the job
- Identify machines, tools, equipment, and technologies used in the job

Job Description

Example of a Job Description

Job Title: Record Clerk **Job No.** 011
Supervisor: Record Supervisor **Job Grand** –III
Supervises: None **Date:** 2/21/12

Job Summary: Originate, process, and maintain comprehensive records; implement required controls; collect and summarize data as requested.

Job Duties and Responsibilities:

- Review a variety of documents, listings, summarizes, etc, for completeness and accuracy.
- Check records against other current sources such as reports or summaries; investigate differences and take required action to ensure that records are accurate and up to date; compile and summarize data report format as required.
- Implement controls or obtaining, preserving, and supplying a variety of information. Prepare simple requisitions, forms, and other routine memoranda.
- Provide functional guidance to lower-level personnel as required.

Working Conditions: Normal working conditions. But visits sites on average twice a week. Eight hours per day

Relationships:

- With equivalent officers in other departments.
- Maintains formal and social contacts with local officials.

Job Characteristics: Skilled operation of computer, calculating machine, or key punch machine is not necessarily a requirement of this job.

The above information is correct and approved by

(Signed) (Signed)
Job Analyst In charge Manager

- A job description is a written document that summarizes the duties, responsibilities, and requirements of a specific job.
- It is derived from the information gathered during job analysis.
- A job description typically includes the job title, a summary of the position, essential duties and responsibilities, qualifications, and sometimes the working conditions and expected outcomes.

Job Specification

- Job specification, also known as employee specification, outlines the qualifications, skills, knowledge, and personal attributes required for an individual to perform a specific job.
- It is derived from the job analysis and is often included as part of the job description.

- Job specifications help in the recruitment and selection process by providing a clear understanding of the ideal candidate's characteristics.
- A shop wants a cashier. The job specification says *basic math skills and computer billing knowledge*. This helps HR select only candidates who meet these requirements.

Key Components of a Job Specification

- **Job Title**: Official role designation (e.g., "Software Engineer").
- **Qualifications**: Required education and certifications (e.g., degree).
- **Experience**: Relevant work or industry experience (e.g., "3-5 years").
- **Skills and Competencies**: Technical, soft, or specialized skills (e.g., Python, communication).
- **Physical and Mental Requirements**: Physical or cognitive demands (e.g., lifting, attention to detail).
- **Personal Attributes**: Desired traits (e.g., adaptability, teamwork).
- **Responsibilities** (Optional): Brief overview of duties.
- **Work Environment**: Conditions, tools, or travel requirements (e.g., remote work, shift-based).

4.1.3 Recruitment and Selection

- **Recruitment** is the process of attracting qualified candidates for job vacancies.
- *Attracting a pool of qualified candidates through internal (promotions) or external sources (job ads, agencies).*

- **Internal Sources:**
  - *promotions, transfers, employee referrals*
- **External Sources:**
  - *job portals, campus recruitment, employment agencies, social media, job fairs*

Selection

- **Selection** is choosing the most suitable candidate from the applicant pool.
- Screening, interviewing, testing, and choosing the best fit via structured processes.

<u>Selection Process</u>

1. Vacancy Announcement
2. CV & Document Screening
3. Written / Technical Test
4. Technical Interview
5. HR / Management Interview
6. Background / Reference Check
7. Appointment Letter

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Human Resource Training (On the Job and Off the Job)

Training is a systematic process of developing employees' knowledge, skills, and attitudes required to perform a specific job effectively.

- Human Resource Training is a **systematic and planned process** of developing employees' skills, knowledge, and attitudes.
- It helps employees **perform their current job more effectively**.
- It aims to **improve productivity, quality, and work efficiency**.
- Training prepares employees for **future roles and responsibilities**.
- It can be provided through **on-the-job** or **off-the-job** methods.

1. On-the-Job Training (OJT)

- **On-the-Job Training (OJT)** is a practical approach to employee development where individuals learn the skills, knowledge, and competencies they need for their role while performing actual work tasks in a real (or simulated) workplace environment.

- *On-the-job training is a method in which employees learn by performing their actual job tasks at the workplace under the guidance of supervisors, seniors, or experienced employees.*

Major Methods of On-the-Job Training:

- **Job Instruction Training (JIT):**
  - ✓ *Step-by-step explanation and demonstration of job tasks.*
  - ✓ ***Example****:* A senior software developer demonstrates how to use Git for version control, then asks the junior developer to commit code following the same steps.
- **Coaching:**
  - ✓ Continuous guidance and feedback provided by supervisors.
  - ✓ ***Example****:* A team lead guides a junior programmer on writing optimized SQL queries during live project work.
- **Mentoring:**
  - ✓ Senior employees guide juniors for long-term career development.
  - ✓ ***Example****:* A senior system architect mentors a fresh IT graduate on system design, coding standards, and career planning.
- **Apprenticeship Training:**
  - ✓ Combination of practical work and theoretical instruction (common in technical jobs).
  - ✓ ***Example****:* A trainee network engineer works under an experienced network administrator to learn server configuration and troubleshooting.
- **Job Rotation:**
  - ✓ Employees are shifted from one job to another to gain broader skills.
  - ✓ ***Example****:* An IT trainee rotates through software development, quality assurance (QA), and IT support to gain multi-skill exposure.

Advantages:

- Low cost and time-saving
- Practical and realistic learning
- Immediate application of skills

Limitations:

- Risk of mistakes affecting work
- Quality depends on the trainer
- May disrupt regular operations

Off-the-Job Training

- **Off-the-Job Training** refers to employee training that is conducted **away from the normal work environment and daily duties**.

- It usually happens in a separate location like a **training center, classroom, conference venue,** or **online platform** and focuses on theoretical knowledge, concepts, and skills not tied to immediate task performance

Major Types of Off-the-Job Training

1. **Classroom Lectures / Workshops**
   Formal, instructor-led sessions covering theories, principles, and concepts.
2. **E-Learning / Online Courses**
   Self-paced or virtual classroom training via learning management systems (e.g., Coursera, LinkedIn Learning, company portals).
3. **Conferences & Seminars**
   Industry events where employees learn from external experts and network with peers.
4. **Simulation Exercises**
   Use of mock setups (e.g., flight simulators, business games, VR training) to mimic real scenarios in a controlled setting.
5. **Role-Playing & Case Studies**
   Interactive methods where trainees solve realistic business problems or act out workplace situations.
6. **University / College Courses**
   Sponsored degree or certification programs relevant to the employee's role or career development.
7. **Outdoor / Adventure Training**
   Team-building activities focused on leadership, communication, and problem-solving in non-work settings.

Pros

- Training delivered by experienced experts
- Well-structured and detailed training methods
- Flexible scheduling for multiple employees
- Career-long access to useful learning resources

Cons

- May disrupt employees’ regular work schedules
- Can be expensive, especially with high turnover
- Risk of low-quality or ineffective training programs
- Training may be wasted on unmotivated employees

Comparison

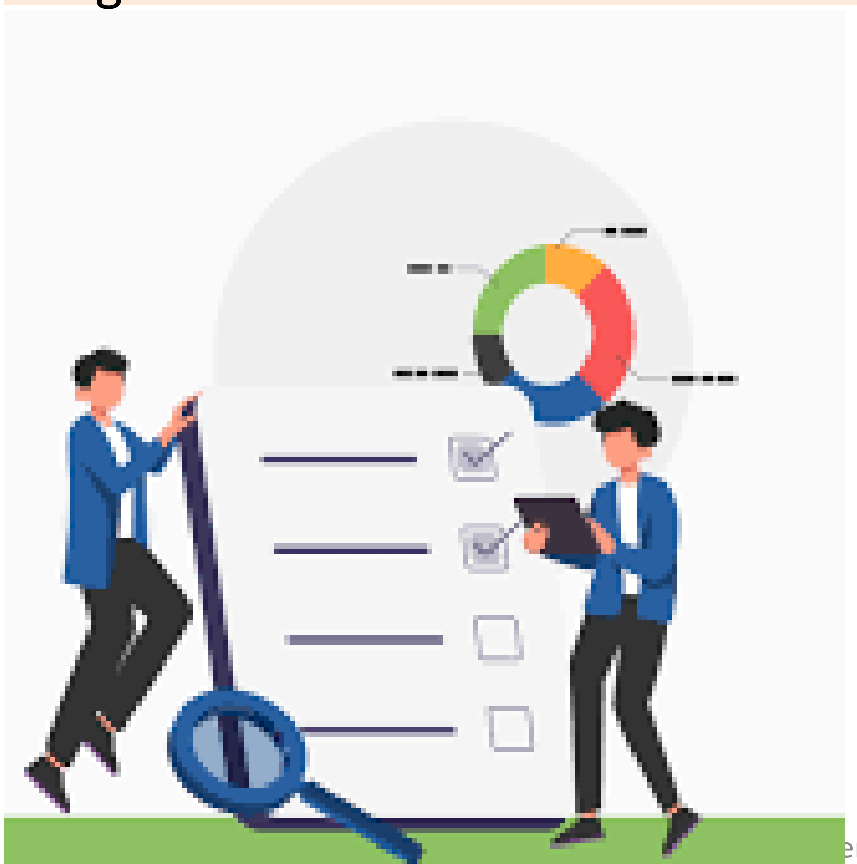
Aspect	Off-the-Job Training	On-the-Job Training
Location	Away from workplace	At the workplace
Focus	Theory, concepts, broad skills	Practical, task-specific skills
Cost	Usually higher (direct costs)	Usually lower (integrated with work)
Application	Delayed	Immediate
Trainer	External expert or specialized instructor	Supervisor, mentor, or experienced colleague
Customization	Often generic	Highly job-specific
Disruption	High (employee is away from job)	Low to moderate (learns while working)

Trends in Training:

- **E-Learning and Digital Platforms**:
  - Use of Learning Management Systems (LMS) like Moodle or TalentLMS for flexible, self-paced learning.
- **Microlearning**:
  - Short, focused training modules for quick skill acquisition.
- **Personalized Training**:
  - AI-driven programs tailored to individual needs.
    - **Gamification**:
      - » Incorporating game elements to increase engagement.
        - **Diversity and Inclusion Training**:
          - Addressing workplace equity and cultural competence.

Performance Appraisal

- Performance appraisal is a systematic process used by organizations to evaluate an employee's job performance and overall contribution to the company.
- Performance appraisal is the **ongoing process of evaluating employee job performance** to understand their contributions and identify areas for growth.



A formal process used by organizations to **evaluate an employee's work performance**, usually to measure how well they are doing their job, give feedback, and decide on promotions, rewards, or training needs.

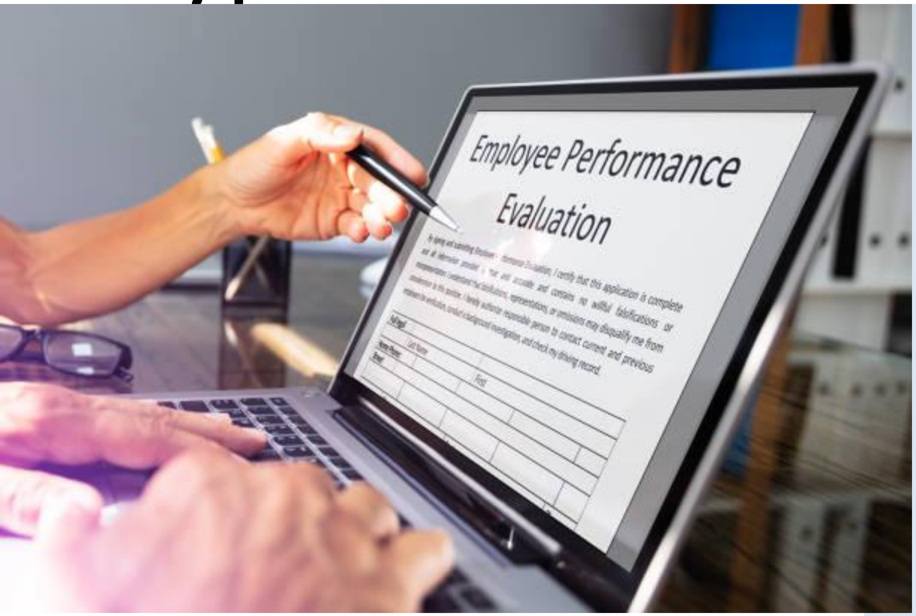
Performance appraisal means evaluating an employee's current or past performance relative to performance standards

\- **Gary Dessler**

Process of Performance Appraisal

1. **Establish Performance Standards**
   - *Set clear, measurable, job-related standards.*
2. **Communicate the Standards**
   - *Make sure employees understand what is expected.*
3. **Measure Actual Performance**
   - *Collect data through observation, reports, and results.*
4. **Compare Actual Performance with Standards**
   - *Identify gaps, strengths, and improvement areas.*
5. **Discuss the Appraisal with the Employee**
   - *Provide feedback, clarify issues, and listen to the employee.*
6. **Initiate Corrective Action**
   - *Provide training, counseling, rewards, or adjustments to improve performance.*

Types of Performance Appraisal



Rating Scales: Employees are assessed using a numerical scale on predefined criteria, indicating performance levels.

360-Degree Feedback: Feedback is collected from peers, supervisors, subordinates, and self-assessment for a well-rounded evaluation.

- **Behavioral Observation:** Observing employees' actions and behaviors to assess job performance and competencies.
- **Management by Objectives (MBO):** Setting specific objectives with employees and evaluating performance based on goal achievement.
- **Critical Incident Technique:** Identifying specific positive and negative events that reflect an employee's performance.

360-Degree Performance Appraisal

- concept began to emerge in the **1980s**.
- Popularized by **Jack Welch** at **General Electric**.
- A comprehensive method that collects feedback from a range of sources, including managers, peers, subordinates, and customers.
- Provides a holistic view of the employee's performance and interpersonal skills.
- This approach provides a well-rounded perspective on an individual's skills, behaviors, and competencies.

Strength

- Ensure anonymity to encourage honest and unbiased feedback
- Use primarily for employee development, not only for performance evaluation
- Provide proper training to raters to improve accuracy and objectivity
- Follow up with coaching and development plans based on feedback results

Challenges

- Time-consuming and costly to design, administer, and analyze
- Risk of turning into a "popularity contest" rather than objective assessment
- Conflicting feedback from different raters may confuse employees
- Cultural resistance, especially in hierarchical organizations like many South Asian/Nepali contexts

limitations of performance appraisal:

- **Judgment Errors**
  - *Rater bias (halo effect, leniency, central tendency) reduces appraisal accuracy*
- **Poor Appraisal Forms**
  - *Unclear, outdated, or poorly designed appraisal tools lead to ineffective evaluation*
- **Lack of Rater Preparedness**
  - *Appraisers are not properly trained or prepared to conduct fair appraisals*
- **Ineffective Organizational Policies and Practices**
  - *Flawed appraisal system design and weak HR policies undermine effectiveness*

Essentials of Performance Appraisal

- ***Reliable Measures:*** *Consistent results.*
- ***Standardization:*** *Same process for all.*
- ***Just and Fair:*** *Unbiased and equitable.*
- ***Viability:*** *Practical and easy.*
- ***Clear Objective:*** *Defined purpose.*
- ***Trained Appraisers:*** *Skilled evaluators.*
- ***Feedback and Participation:*** *Communication and input.*
- ***Confidential:*** *Private information.*

4.1.6 Challenges in Managing People in ICT Workforce

- **Skills Gaps and Rapid Obsolescence**:
  - Constant need for upskilling in emerging technologies like AI and cybersecurity.
- **Talent Shortage and High Turnover:**
  - Competitive market leads to poaching and job-hopping.
- **Remote/Hybrid Work Management:**
  - Ensuring productivity, collaboration, and security in distributed teams.
- **Burnout and Work-Life Balance:**
  - High-pressure deadlines and on-call demands.
- **Diversity and Inclusion:**
  - Attracting underrepresented groups in a traditionally male-dominated field.
- **Attracting/Retaining Talent:**
  - Need for competitive compensation, growth opportunities, and innovative cultures.

Meaning of control

Control is the **process of monitoring performance, comparing it with standards, and taking corrective action** to achieve organizational goals.

- Controlling is the process of monitoring activities to ensure that they are being accomplished as planned and of correcting any significant deviations.
- Controlling is the process of setting standard for output, measuring actual output, finding deviation if any, and initiating corrective actions to eliminate or at least minimize the deviation between predetermined output and actual output.

- *It is the only management function* ***which ensures the activities are at the right track to attain organizational goals****.*

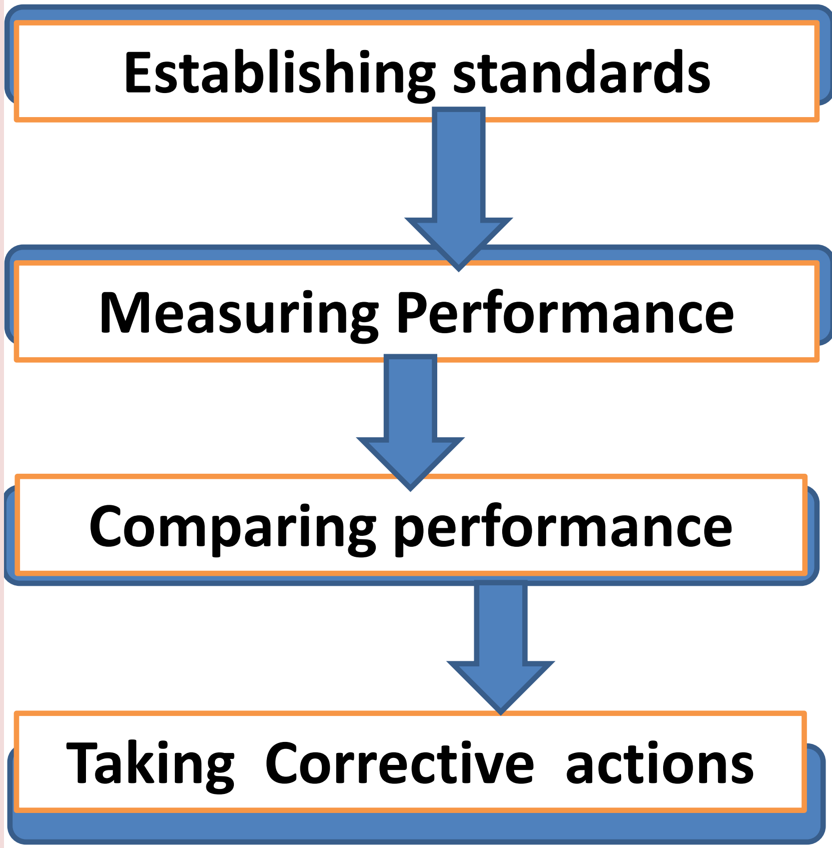
Importance of Control

- *Ensures* ***goal achievement*** *and organizational efficiency.*
- *Minimizes* ***errors and deviations*** *in processes.*

- *Enhances* ***coordination among departments****.*
- *Facilitates* ***decision-making and corrective actions****.*
- *Builds* ***accountability and transparency*** *in organizations.*

Process of Controlling

- **Setting Standards**
  - ✓ Fixing expected performance levels in advance.
  - ✓ **Eg**. Software response time must be within 2 seconds.
- **Measuring Performance**
  - ✓ Collecting data on actual system performance.
  - ✓ **Eg**. Load time measured using performance testing tools.
- **Comparing with Standards**
  - ✓ Comparing actual performance with predefined standards.
  - ✓ **Eg**. Actual load time 3.5 seconds vs standard 2 seconds.
- **Taking Corrective Action**
  - ✓ Taking steps to correct deviations from standards.
  - ✓ **Eg**. Code optimization and bug fixing are performed.



Types of control

1. Feedforward (Preliminary):

- Prevents issues before they occur
- **e.g.** training, policies.

2. Concurrent (Real-Time):

- Monitors ongoing activities
- **e.g.** quality checks

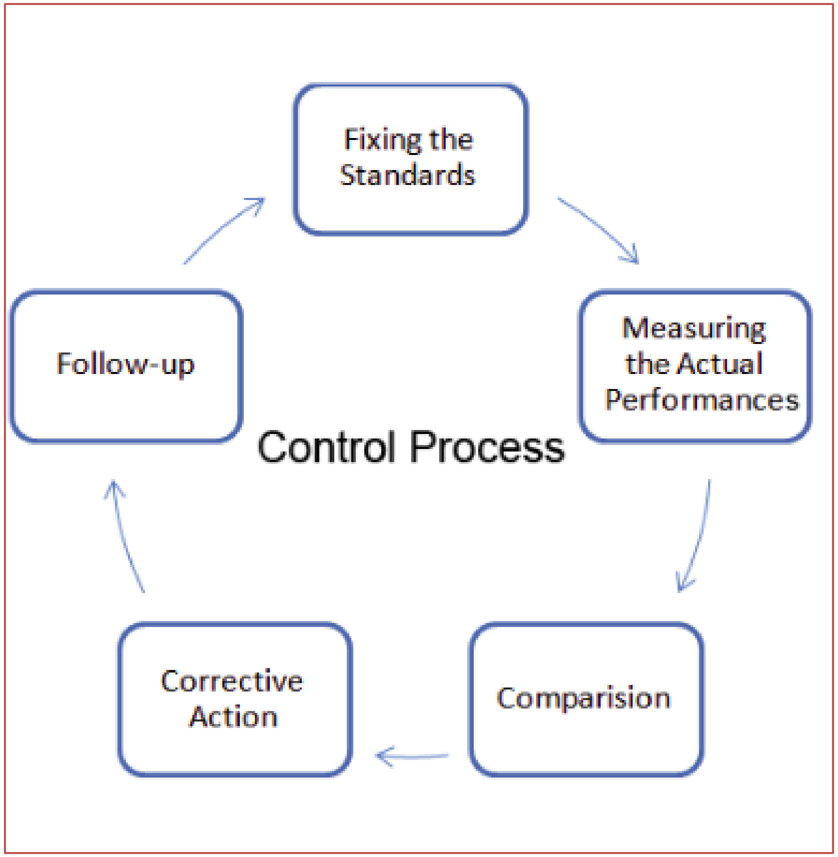
3. Feedback (Post-Action):

- Evaluates after completion
- **e.g.** performance reviews

4.2.3 Techniques for Control

- **<u>Traditional:</u>**
  - *Budgetary control,*
  - *Break-even analysis,*
  - *Statistical reports.*
- **<u>Modern:</u>**
  - *PERT (Program Evaluation and Review Technique)*
  - *CPM(Critical Path Method) for project scheduling.*

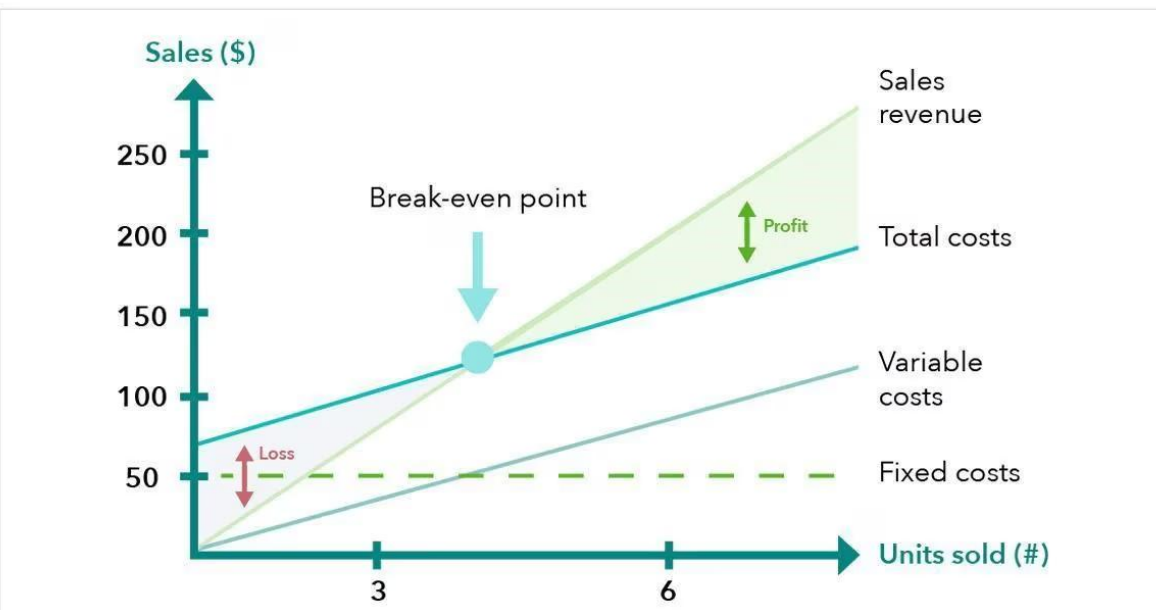
Budgetary Control



- Monitoring costs against the planned budget to ensure financial discipline.
- Compare actual expenditure vs budgeted expenditure and analyze variances.
- **Eg.** *In software projects, monitoring cloud hosting costs or software license fees against allocated budget.*

Break-even Analysis / Cost Control

- Determine the point at which costs and revenues are equal to plan resource allocation.



- **Eg.** *In software projects, determine how many software licenses or services need to be sold to cover development cost.*

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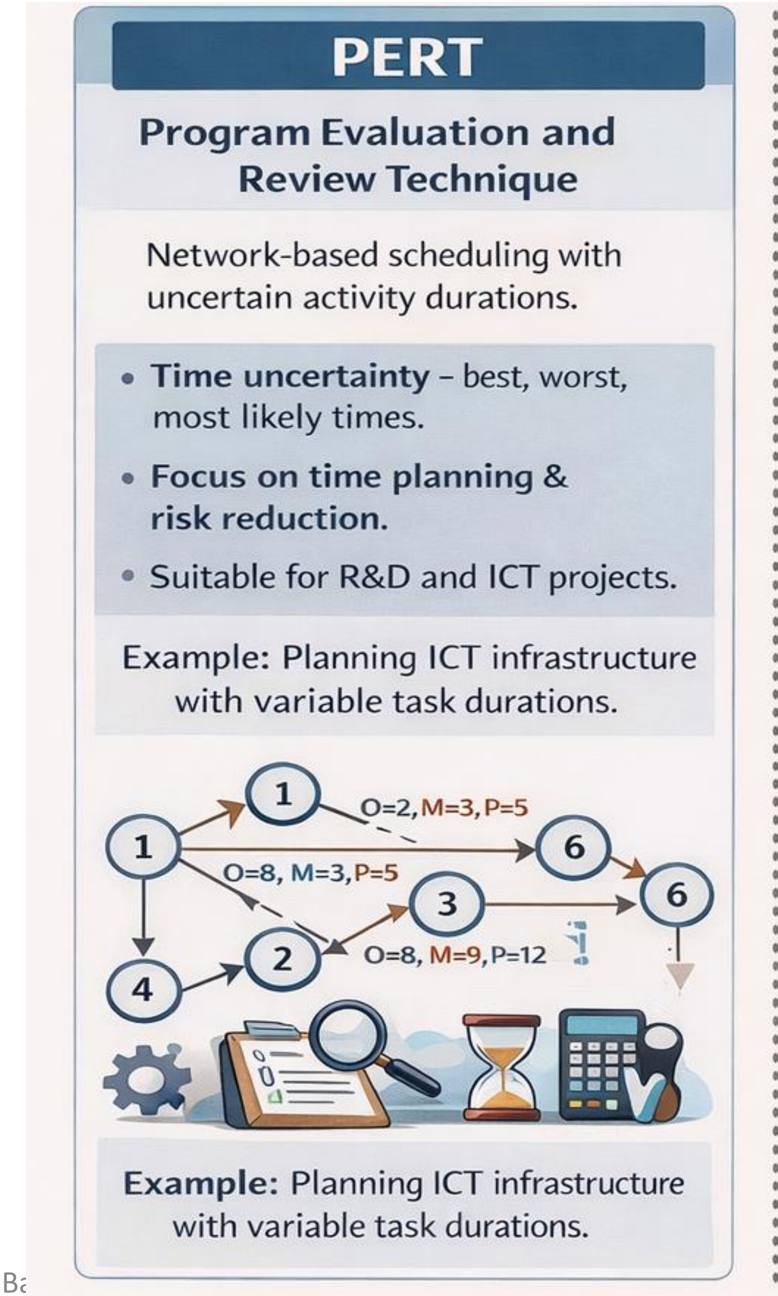
Statistical Reports / Statistical Control

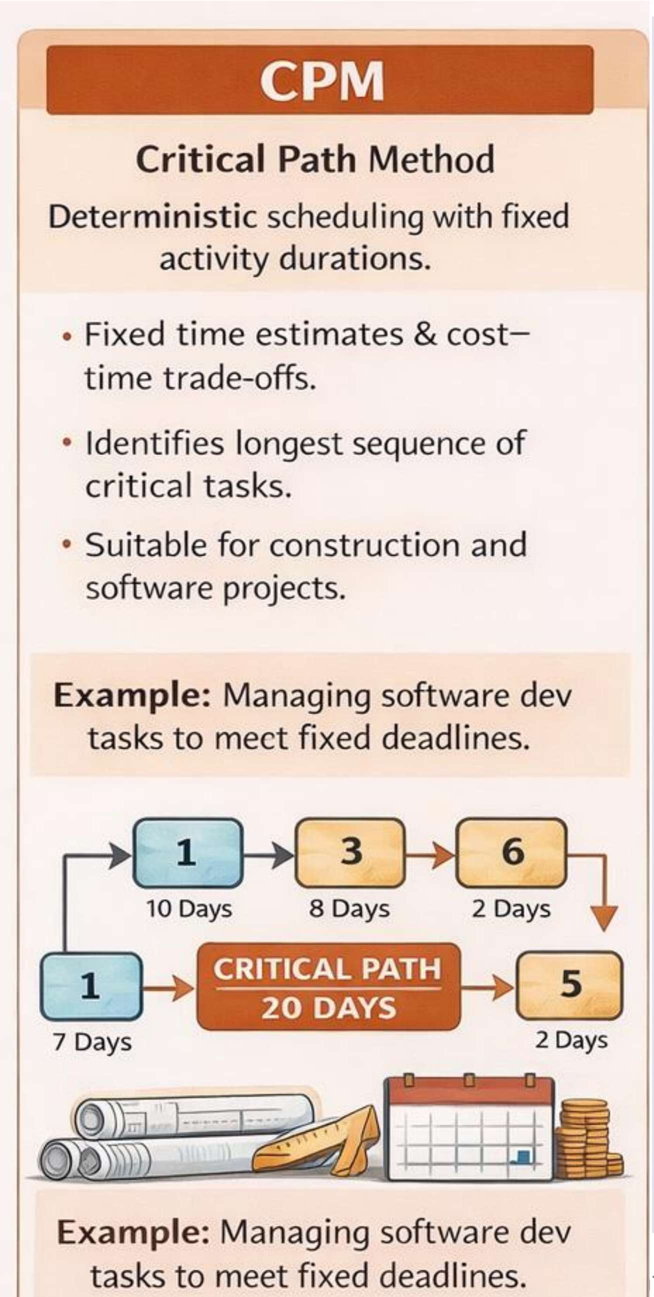
- Using statistical tools and data to monitor performance and quality.
- Mean, standard deviation, control charts, Six Sigma.
- **Eg.** *Software development: Number of bugs per 1000 lines of code; detect if error rate exceeds target.*

PERT (Program Evaluation and Review Technique)

- A network-based project scheduling and control technique.
- Used when activity time is uncertain or variable.
- Focuses on **time planning**, **coordination**, and **risk reduction**.
- Helps managers identify **earliest and latest start times** of activities.
- Supports decision-making in large and complex engineering projects.
- **Example:** *Planning an ICT infrastructure project where task durations are uncertain.*



CPM (Critical Path Method)



- A deterministic project scheduling technique with fixed activity time.
- Identifies the **critical path** (longest sequence of activities).
- Any delay in critical activities delays the entire project.
- Helps in **resource allocation** and **time optimization**.
- Useful for monitoring project progress and controlling deadlines.
- **Example:** *Managing software development tasks where delivery dates are fixed.*

4.2.4 ICT Tools for Effective Control of Engineering Projects and Organizations

1. *Project Management Software*
2. *Collaboration & Communication Platforms*
3. *Document Management & Version Control Systems*
4. Building Information Modeling(*BIM) & Design Collaboration Tools*
5. Enterprise Resource Planning(*ERP) & Portfolio Management Tools*
6. *AI-Enhanced & Advanced Tools*

1. Project Management Software

- Plans and schedules engineering activities and tasks
- Allocates and controls time, cost, and resources
- Tracks progress and milestones for timely completion
- **Examples:**
  - *Microsoft Project*
  - *Primavera P6*
  - *Asana*

2. Collaboration & Communication Platforms

- These platforms enable real-time communication and teamwork among project stakeholders.
- They support messaging, video meetings, file sharing, and integration with other tools.
- Such platforms improve coordination, reduce delays, and enhance team productivity.
- **Examples:** *Microsoft Teams, Slack, Zoom*

3. Document Management & Version Control Systems

- These systems store, organize, and manage project documents in a centralized location.
- They track document versions, control access, and maintain audit trails.
- This ensures accuracy, security, and consistency of project information.
- **Examples:** *SharePoint, Google Drive, GitHub*

4. Building Information Modeling (BIM) & Design Collaboration Tools

- BIM tools create digital representations of physical and functional characteristics of projects.
- They support 3D modeling, clash detection, and design coordination among stakeholders.
- These tools reduce errors, rework, and improve construction efficiency.
- **Examples:** *Autodesk Revit, Navisworks, BIM 360*

5. Enterprise Resource Planning (ERP) & Portfolio Management Tools

- ERP and portfolio tools integrate finance, HR, procurement, and project management functions.
- They provide a unified view of resources, costs, and multiple projects.
- These systems support strategic decision-making and organizational control.
- **Examples:** *SAP ERP, Oracle ERP, Microsoft Dynamics 365*

6. AI-Enhanced & Advanced Tools

- AI-enhanced tools use artificial intelligence to automate and optimize project activities.
- They assist in risk prediction, intelligent scheduling, and performance analysis.
- Such tools improve accuracy, efficiency, and proactive decision-making.
- **Examples:**
  - *AI-based scheduling tools*
  - *Generative design software*
  - *AI-powered document analysis tools*

Characteristics of Effective Control System

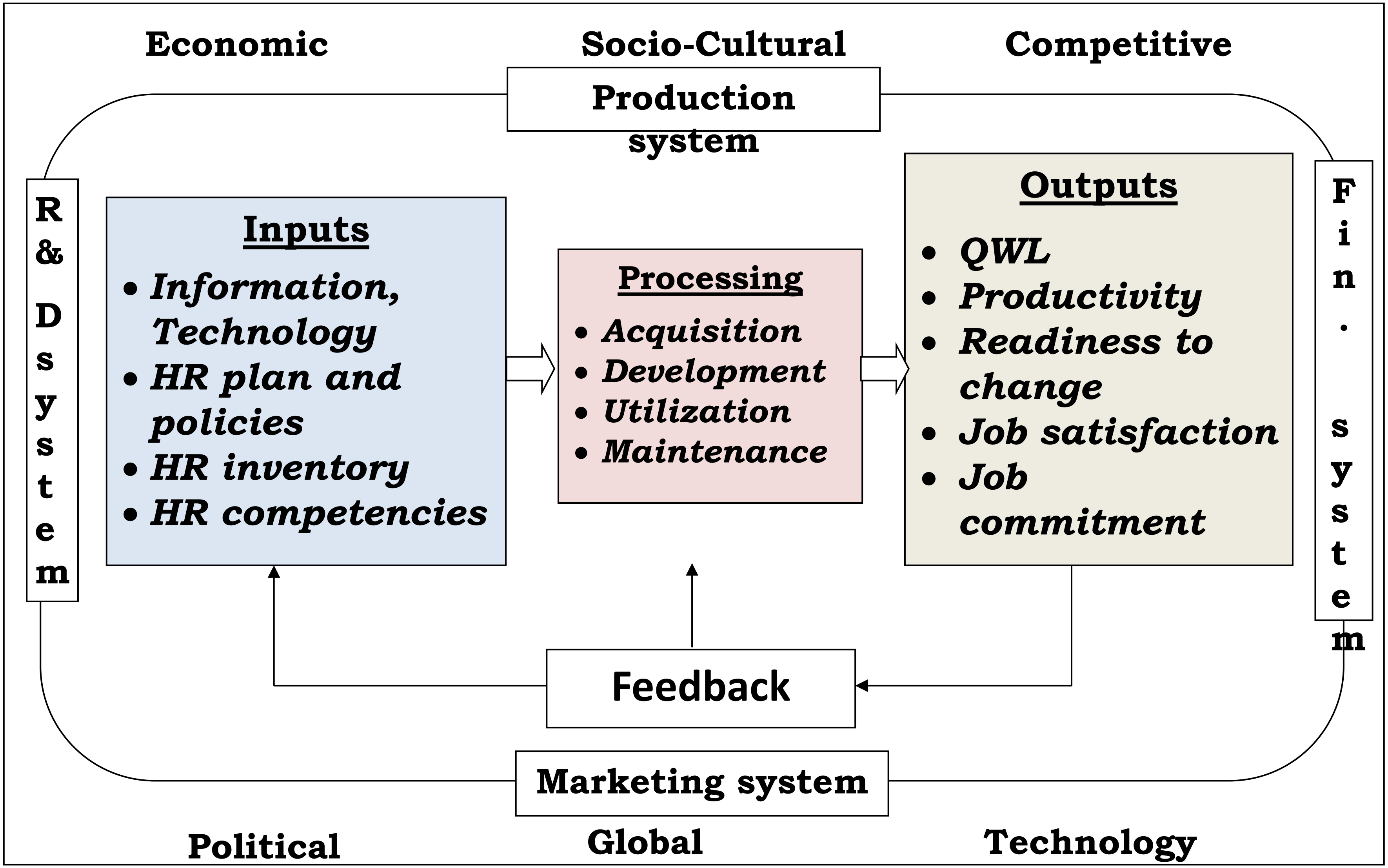
- ***Suitability***
- ***Simplicity***
- ***Economically realistic***
- ***Integrated***
- ***Flexible***
- ***Objectivity***
- ***Accuracy***
- ***Strategic Focus***
- ***Acceptable***
- ***Quick***
- ***Corrective***
- ***Forward looking***

Importance of Controlling

- *Implementation of plan*
- *Improvement in efficiency*
- *Basis of future action*
- *Aid to decentralization*
- *Morale of employees*
- *Means of coordination*
- *Effective supervision*
- *Maximizing productivity*

Extra Notes

Human Resource Management System



Components of HRM

Acquisition Function

Development Function

Utilization Function

Maintenance Function